

ENTIRE EDIH One Day Training Course on AI Based Scheduling

Helmut Simonis
School of Computer Science and Information Technology
University College Cork
Cork, Ireland
email:helmut.simonis@insight-centre.org

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Abstract

This document describes the detailed structure of a one day training course on AI based scheduling for the ENTIRE EDIH project. This is a cut-down version of the existing two-day course to fit into a single day. The training course provides an overview of current Constraint-based techniques to model and solve scheduling problems arising in project scheduling and manufacturing industries. It provides some hands-on experience with available open-source tools, and gives examples of SME based scheduling solutions.

1 Introduction

This document provides an outline of a one-day training program on AI based scheduling using Constraint Programming techniques. The document details the topics covered from the viewpoint of the course developer, it is not intended to be read by the potential audience. A separate description of the course focussing on business needs and benefits will be developed in parallel.

2 Intended Audience

The course is intended for decision makers and persons facing scheduling problems in their professional work. IT personnel or consultants charged with choice of tools, development of models, or integration into the existing infrastructure will also be interested.

3 Learning Outcomes

After attending the course, the students should be able to

- Position Constraint Based Scheduling in the general AI field
- Understand the need and potential for automated scheduling

- Understand the basic ideas behind Constraint-Based Scheduling
- Identify and apply the basic constraint types occurring in scheduling
- Understand the strength and weaknesses of available tools
- Be aware of basic visualization methods for scheduling

4 Timetable

Table 1 shows the format of the course delivered in a two-day event, with ten hours total teaching time. The course consists of seven modules, which will be described in more detail in Section 5. Alternatively, the course could be presented as ten one hour sessions (plus exercises) for self-study.

Table 1: Proposed Two Day Course Structure

Time	Topic
09:00-09:30	Registration and Welcome
09:30-10:30	Introduction & Motivation
10:30-11:00	Coffee
11:00-12:30	Scheduling Concepts
12:30-13:30	Lunch
13:30-15:00	Case studies
15:00-15:30	Coffee
15:30-16:30	Hands-on Experiments

5 Modules

This section gives a brief bullet-point list of the key elements of each of the modules of the course. The course will be delivered as a mix of overview slides, some example problems and solutions using different solvers, and a hands-on session to explore programs and demonstrators.

The overall approach is morphology based, i.e. based on the structure of problems found in industry. This differs from a complexity/structural analysis often used in OR, or a technology based presentation used in Computer Science.

5.1 Introduction & Motivation

- AI - More than Chat-GPT
 - Deductive vs stochastic approaches
- What is scheduling?
- What is Constraint-Based Scheduling?
- Solution approaches
- Motivating examples

- From CP and CPAIOR papers
 - Taken up again in case studies section
- Benefits of automated scheduling
- Why not just use a package?
- What is not covered?
 - Setup and transport times
 - Alternative process paths
 - Stochastic variants
 - Process and semi-process based manufacturing
 - Internal operations of a constraint solver

5.2 Scheduling Concepts

- Core concepts: Products, orders, jobs, tasks
- Temporal relations
 - End-to-start precedence
 - Precedence graph (linear, tree)
 - Max waiting time
 - Intra-job vs inter-job
- Resource Constraints
 - Disjunctive resources
 - Cumulative resources
 - Machine Choice
- Release date and due date (hard, soft)
- Problem classification
 - RCPSP
 - Job-Shop, flow-shop, open-shop
- Visualization methods
 - Gantt chart (job/machine view)
 - Precedence graphs
 - Process diagrams
 - Resource profiles

5.3 Case Studies

- Industrial success stories from SMEs
- Show how they fit into the framework provided
- Test Scheduling (ABB)
- No-wait flowshop
 - Difficult feasible solutions
 - Difficult analysis of solution quality
- Print-shop capacity planning (Kostwein)
 - Fielded application from Austria
- Yacht refit scheduling
- Radiation therapy patient scheduling
- Disaster Planning: Non-preemptive Evacuation Scheduling
- Feedmill Scheduling (Dalgety)
 - Short term production scheduling
 - Storage capacity limits
 - Ability to promise

5.4 Experiments

- Hands-on experience with the Insight Scheduling Tool
 - Open-access tool using different back-end solvers
 - Interactive use
 - Installed on our machines
 - Packages available for installation of student's machines (Windows/Mac)
- Based on provided examples
- Visualization of results

6 Conclusion